Cut for derivability: derivations compose by concatenation

The Open Logic Project contributors

openlogic:first-order-logic/axiomatic-deduction:local-snapshot-20261013; CC BY 4.0

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Open Logic Project unit file sources/olp-first-order-logic-axiomatic-deduction/pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the follow-up paragraph generalising to A_1..A_n is outside the span and not claimed by the card; the display of the spliced sequence is verbatim.

Editorial note. The label prop:transitivity names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own prop environment in the unit source.

Allowed standard prerequisites (editorial).

- Definition of derivation from assumptions. Earlier in the same chapter: a derivation is a finite sequence whose lines are axioms, assumptions, or rule consequences. The spliced sequence is checked against this clause-by-clause justification requirement at lines 875-877.

Proposition 0.1 (Transitivity)**.** *If* $\Gamma \vdash !A$ *and* $\{!A\} \cup \Delta \vdash !B$*, then* $\Gamma \cup \Delta \vdash !B$*.*

Proof. Suppose $\{!A\} \cup \Delta \vdash !B$. Then there is !!aderivation $!B_1$, ..., $!B_l = !B$ from $\{!A\} \cup \Delta$. Some of the steps in that !!derivation will be correct because of a rule which refers to a prior line $!B_i = !A$. By hypothesis, there is !!aderivation of $!A$ from Γ, i.e., !!aderivation $!A_1$, ..., $!A_k = !A$ where every $!A_i$ is an axiom, !!aelement of Γ, or correct by a rule of inference. Now consider the sequence

$$!A_1, \ldots, !A_k = !A, !B_1, \ldots, !B_l = !B.$$

This is a correct !!derivation of $!B$ from $\Gamma \cup \Delta$ since every $B_i = !A$ is now justified by the same rule which justifies $!A_k = !A$. □